

Customer Churn Prediction & Model Evaluation

End-to-End Data Science Pipeline Documentation & Strategic Insights

1. End-to-End Pipeline & Exploratory Data Analysis

This documentation outlines the full workflow executed in the Jupyter Notebook, spanning raw data cleaning, feature engineering, exploratory data analysis, data scaling, model development, and final evaluation.

Data Preprocessing & Feature Engineering

- **Data Cleaning:** Handled missing values and string-encoded numerical features (e.g., converting TotalCharges to numeric and resolving whitespace anomalies). Binary categorical columns were mapped to standard numerical flags.
- **Feature Creation (Charge-per-Tenure):** Engineered a composite ratio feature defined as $\text{MonthlyCharges} / (\text{Tenure} + 1)$. This engineered feature directly captures customer price-density over time and acts as a primary indicator of cost sensitivity.
- **Add-on Services Aggregation:** Engineered an explicit total count feature by aggregating individual optional services including Online Security, Online Backup, Device Protection, Tech Support, Streaming TV, and Streaming Movies.

Key EDA Insights

- **Add-on Services vs Churn Impact:** Customers with multiple add-on services exhibit significantly lower churn. Subscribers with 0 add-ons represent the highest risk group, reaching nearly 50% churn due to low product integration and minimal switching friction.
- **Charge-per-Tenure Behavior:** Churned users display a drastically higher charge-per-tenure ratio (~ 12.0) compared to retained customers (~ 3.6). This demonstrates that early-tenure customers facing high monthly costs perceive a poor cost-to-value ratio and depart quickly.

Business Action Plan:

To mitigate high early-tenure churn, target customers with 0–1 add-ons with tailored promotional bundles (e.g., free 3-month trial of Tech Support or Online Backup). Moving a customer from 0 to 1 add-on service reduces their probability of churn from 44% down to 21%.

2. Preprocessing & Data Scaling Methodology

Train-Test Leakage Prevention

- **Strict Train-Set Fitting:** To maintain robust model validation, all scalers and transformations were fit strictly on the training set (X_{train}) and subsequently applied to the test set (X_{test}).

- **Theoretical Justification:** Fitting preprocessing tools on the full dataset or test set introduces severe data leakage. Test-set parameters (mean, standard deviation) leak into model training, producing artificially inflated, ungeneralizable performance metrics.

Scaler Selection Analysis

- **StandardScaler Evaluation:** StandardScaler standardizes features by removing the mean and scaling to unit variance (z-scores). It demonstrated superior stability across distance-based algorithms (KNN, SVM) and gradient-based models (Logistic Regression).
- **MinMaxScaler Comparison:** MinMaxScaler scales features into a fixed range [0, 1]. While effective for bounded neural networks, it suffers when outliers exist in financial ratios like TotalCharges or Charge-per-Tenure, making StandardScaler the superior final choice.

3. Model Benchmarking & Experimental Results

Multiple classification models were trained and systematically evaluated on the test dataset using standard classification metrics: Accuracy, Precision, Recall, F1-Score, and ROC-AUC.

Model Name	Accuracy	Precision	Recall	F1-Score	ROC-AUC
Logistic Regression	0.806	0.658	0.534	0.589	0.846
Gradient Boosting	0.801	0.643	0.528	0.580	0.845
Random Forest	0.789	0.612	0.495	0.547	0.823
Support Vector Machine (SVM)	0.795	0.630	0.508	0.562	0.801
K-Nearest Neighbors (KNN)	0.768	0.552	0.480	0.513	0.742
Naive Bayes	0.692	0.448	0.831	0.583	0.828

4. Model Selection Rationale & Trade-off Analysis

 Primary Recommendation: Logistic Regression

- **Peak ROC-AUC Score (0.846):** Logistic Regression achieved the highest overall ranking capacity across all thresholds, effectively separating potential churners from stable subscribers.
- **Top Accuracy & F1-Score:** Delivers the highest accuracy (80.6%) and a balanced F1-Score (0.589), outperforming ensemble models like Random Forest while maintaining full linear interpretability.
- **Computational Efficiency:** Compared to Gradient Boosting (ROC-AUC 0.845), Logistic Regression requires significantly lower compute resources, zero hyperparameter tuning latency, and enables direct coefficient extraction for business reporting.

🔍 Strategic Model Choice & Business Trade-offs:

- High-Recall Objective (Naive Bayes): Naive Bayes achieved the highest Recall (0.831), identifying 83.1% of all churned customers. However, its low Precision (0.448) results in high false alarms.
- Business Strategy Alignment: Choose Naive Bayes if customer retention outreach costs are extremely low and missing a churner is expensive. Choose Logistic Regression if marketing budgets are constrained and targeted precision is required.